

SARVESH PRAJAPATI

✉ prajapati.s@northeastern.edu  [Sarvesh Prajapati](#)  [prajapatisarvesh](#)  [prajapatisarvesh](#)

EDUCATION

- PhD in Electrical Engineering and Computer Engineering**, Northeastern University Dec 2028
Focus: *Robotics*, and *Autonomous Systems*.
- M.S. in Robotics**, Northeastern University, GPA: 3.75/4.00 Dec 2024
Thesis: *Physics-Informed Motion Planning Using RGB-Based Spectral Analysis*.
Focus: *Algorithms*, *Reinforcement Learning*, *Robotic Systems*, *Controls*, *Manipulation*, and *Perception*.
- B.E. in Computer Engineering**, Gujarat Technological University, GPA: 3.68/4.00 Jun 2022
Thesis: *Advancements in autonomous mobile robots for warehouse management and small scale industry*.
Focus: *Robotics*, *DBMS*, *Data Structure and Algorithms*, *Operating Systems*, and *Object Oriented Design*.

PUBLICATIONS

Journal Papers

1. A. Trivedi, **S. Prajapati**, M. Zolotas, M. Everett, T. Padir, “Chance-Constrained Convex MPC for Robust Quadruped Locomotion Under Parametric and Additive Uncertainties,” 2025, *Robotics and Automation Letters (RA-L)*.

Peer-Reviewed Conference Papers

1. **S. Prajapati**, A. Trivedi, N. Hanson, B. Maxwell, T. Padir, “Spectral Signature Mapping from RGB Imagery for Terrain-Aware Navigation,” *IEEE Robotic Computing and Communication (IRC)*, Naples, Italy, 2025.
Best Paper Award.
2. A. Trivedi, **S. Prajapati**, A. Shirgaonkar, M. Zolotas, T. Padir, “Data-Driven Sampling Based Stochastic MPC for Skid-Steer Mobile Robot Navigation,” 2025 *IEEE Conference on Robotics and Automation (ICRA)*, Atlanta, GA, USA, 2025.
3. N. U. Akmandor, **S. Prajapati ***, M. Zolotas *, T. Padir, “Re4MPC: Reactive Nonlinear MPC for Multi-model Motion Planning via Deep Reinforcement Learning,” 2025 *IEEE Conference on Automation Science and Engineering (CASE)*, Los Angeles, CA, USA, 2025.
4. N. Hanson *, **S. Prajapati ***, J. Tukpah, Y. Mewada, and T. Padir, “Automated Forest Biomass Mapping with Autonomous Hyperspectral Imaging for Wildfire Monitoring,” 2024 *IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, New York, US, 2024.
5. A. Trivedi, M. Zolotas, A. Abbas, **S. Prajapati**, S. Bazzi, and T. Padir, “A Probabilistic Motion Model for Skid-Steer Wheeled Mobile Robot Navigation on Off-Road Terrains,” 2024 *IEEE Conference on Robotics and Automation (ICRA)*, Yokohama, JP, 2024.
6. Y. Qian, **S. Prajapati**, A. Schwartz, A. Jung, U. Seitz, J. Alfen, L. Lewis, M. Kim, A. Kramer, L. Chukoskie, “Integrated Aerobic Exercise into Adult Second Language Learning in Virtual Reality Game,” 2023 *IEEE Conference on Games (CoG)*, Boston, MA, USA, 2023.

Workshop Papers

1. **S. Prajapati**, A. Trivedi, B. Maxwell, and T. Padir, “Predictive Mapping of Spectral Signatures from RGB Imagery for Off-Road Terrain Analysis,” *Workshop on Resilient Off-road Autonomy*, ICRA, Yokohama, JP, 2024.
2. A. Trivedi, **S. Prajapati**, M. Zolotas, T. Padir, “Online Refinement of Uncertainty Sets for Robust MPC of Quadrupedal Robots Using Convex Cone Programming,” *Workshop on Advancements in Trajectory Optimization and Model Predictive Control*, ICRA, Yokohama, JP, 2024.

*equal contribution

RESEARCH EXPERIENCE

- Research Assistant** Jan 2023 – Present
RIVeR Lab, Northeastern University Boston, US
- Researching on deep learning-based methods to map RGB to hyperspectral images for material recognition and friction estimation enhancing the traversability for mobile and legged robots.
 - Investigating control systems, perception, and learning algorithms, contributing to design of autonomous systems, resulting in 4 published papers in top-tier conferences/workshops and 3 papers under review.
- Applied Scientist Co-Op** Feb 2025 – Dec 2025
Amazon Robotics Westborough, US
- Researching in Innovation Lab, NDA bound.
- Applied Scientist Co-Op** Jan 2024 – Jun 2024
Amazon Robotics Westborough, US
- Created a robust simulation environment from scratch; tested and integrated planning algorithms for robotic manipulators, enhancing system performance and reliability.
 - Conducted research involving Manipulators, Deep Learning, Human-Robot Interaction, Perception, LLM and Optimization in Innovation Lab.
- Research Assistant** Sep 2022 – Dec 2022
ReGame-XR Lab, Northeastern University Boston, US
- Researched on XR game development and cognition for secondary language learning in older adults.
 - Led software development team for **ExerBike project**; modified GTA V, interfaced ANT+ sensor, integrated VR using .NET framework that helped securing \$50k CBH award and conference publication.
- Research Scholar** Jun 2020 – Aug 2022
Gujarat Technological University Ahmedabad, IN
- Researched on controls, embedded systems, motion generation and path-planning for mobile robots. Migrated robots controller from Arduinos to STM32 and wrote a primer on STM32 for undergrads.

TEACHING EXPERIENCE

- Advanced Perception**, Northeastern University Sep 2024 – Dec 2024
Teaching Assistant Boston, US
- Responsible for grading students' weekly research paper reviews and presentations, offering office hours for guidance, and assisting in developing novel aspects for research-based projects.
- Embedded Design Enabling Robotics**, Northeastern University Sep 2023 – Dec 2023
Teaching Assistant Boston, US
- Conducted lab sessions for 100+ students using the De1SoC FPGA board, guiding them through assignments from basic logic gate design to programming a robotic arm.
- GTU Robotics Club**, Gujarat Technological University Jul 2021 – Jul 2022
Team Leader Ahmedabad, IN
- Taught 40 peers robot programming, computer vision and machine learning over a course of 1 year.

HONORS AND AWARDS

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| 2023 | CS Research Mentorship Program Scholar , Google | Boston, US |
| 2022 | Robotics team received best design award and \$1000 prize , DD Robocon | Delhi, IN |
| 2021 | Team represented India in international robotics competition , ABU Robocon | Jimo, CN |
| 2021 | Code-Decode champion , Parul University | Vadodara, IN |
| 2020 | Won CTF and OSCP voucher , Secarmy | Ahmedabad, IN |

SERVICE AND VOLUNTEERING

1. Reviewer for IEEE conferences (ICML, ICRA, SSRR) and Amazon internal conferences.
2. Mentored undergraduates, guided them in robotics research and projects.

SKILLS

Programming Languages	:	C++, Python, MATLAB, JavaScript, Bash
Software/Tools	:	ROS 2, Linux, CMake, Docker, MoveIt!, Git, Gazebo, Nvidia Isaac Sim, AT X
Libraries	:	PCL, OpenCV, PyTorch, TensorFlow, NumPy, Matplotlib, scikit-learn, OpenAI Gym, JAX, Eigen, Open3D, OctoMap, SciPy, Drake, CasADi, CVXPY, Numba
Platforms	:	Clearpath, Universal Robots, Franka Emika, Unitree, Human Support Robot, Xarm, Kinova, TurtleBot, Nvidia Jetson, Raspberry Pi, Arduino, STM32
Interests	:	Playing aerospace and aviation simulation, Piano, Guitar, Books